dewSMART™ Model DSP-Rm

Rechargeable Portable Dewpoint Meter

CE

Features :-

- Mains Rechargeable Operation
- Overall Range -110°C to +20°C Dewpoint
- User Friendly - Simple Operation
- Portable - Fully Self Contained
- Self-Diagnostics on Start-up
- Digital Indication
- Automatic Switch Off
- Quick Connect SS Coupling
- Ordinary "C" Type Rechargeable Batteries
- Excess of 200 Hours of Continuous Operation
- Desiccant Dry-Down Chamber For Rapid Measurements
- Selectable Engineering Units in °C, °F, ppb(v), ppm(v), ppm(w), g/m³ & lb/MMSCF



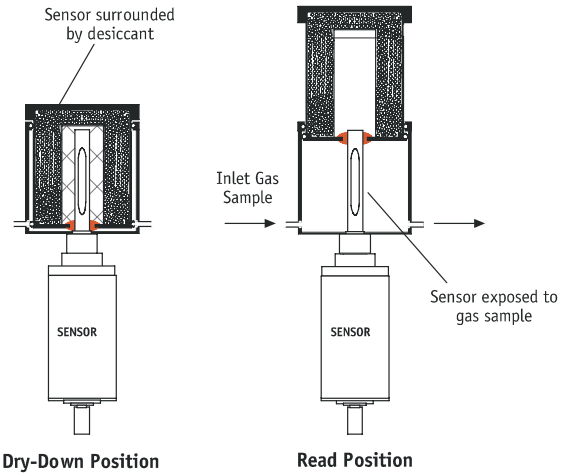


Model DSP-Rm is a fully self-contained portable hygrometer from Alpha Moisture Systems which will deliver the most dependable moisture measurement in industrial and laboratory applications. Designed with the operator in mind, *Model DSP-Rm* is extremely easy to use and the digital *dewSMART*™ technology ensures accurate and reliable readings over long periods with little or no maintenance. The incorporation of stainless steel quick connect fittings, together with the Desiccant Dry-down Chamber, allows rapid measurements which saves both time and cost.

Model DSP-Rm can operate form a mains power supply or standard rechargeable batteries. Designed to operate at atmospheric pressure, *Model DSP-Rm* is available in a total of six different ranges, encompassing an overall range from -110°C to +20°C dewpoint, with the display selectable in various engineering units:- °C, °F, ppm(v), ppb(v), ppm(w), g/m³ & lb/MMSCF. Parts per million (weight) can be programmed for gases other than those pre-programmed (Air, Ar, H_2, SF_6, CO_2 & N_2) by simply inputting the molecular weight of the gas under test.

Full interchangeability of the *dewSMART*™ Sensor ensures guaranteed accuracy and reliability without the need to return the whole unit back to base for calibration and setup.

Model DSP-Rm is supplied ready for use with batteries, battery charger, calibration certificate traceable to National and International Humidity standards, two metres of stainless steel braided PTFE sampling hose, instruction manual and an optional robust transit case.



To allow rapid spot checks it is essential that the measuring sensor is kept drier than the sample to be measured.
The desiccant dry-down chamber is designed to do just this by keeping the sensor surrounded by desiccant when the instrument is not in use, only exposing it to the sample gas when taking measurements. At no stage is the sensor allowed to come into contact with ambient air.

alphamoisture SYSTEMS

Alpha House, 96 City Road
Bradford BD8 8ES England
Telephone: 01274 733100
Facsimile: 01274 733200
E-mail: info@amsystems.co.uk
Web: www.amsystems.co.uk

SENSOR TYPE : Model DDSS - Aluminium Oxide Ultra High Capacitance Digital Sensor

RANGE : Six different ranges encompassing an overall range of -110°C to +20°C dewpoint (-148 to +68°F)
Corresponding ranges in ppm(v), ppb(v), ppm(w), g/m^3 & lb/MMSCF.
Parts per million (weight) programmable for different gases.

DISPLAY : Industrial specification, 2 x 20 Character LCD with programmable bright & dark contrast.

DISPLAY RESOLUTION : 0.1°C dewpoint / 0.1ppm

POWER SUPPLY : 9V DC - Six "C" type rechargeable batteries. Battery charger included.

BATTERY LIFE : In excess of 200 hours during continuos operation.

ELECTRONICS ACCURACY : Better than ± 1% of range.

WARM UP TIME : 15 Seconds

SENSOR CALIBRATION ACCURACY : Better than ±2°C dewpoint. All units supplied with certificates documenting factory calibration against known moisture levels traceable to National & International humidity Standards.

REPEATABILITY : ± 0.1°C dewpoint.

TYPICAL RESPONSE TIME : 95% of reading within 20 seconds in normal operation.

OPERATING CONDITIONS

Temperature : -20°C to +50°C
Humidity : 0-98% RH, NOn-condensing
Storage Temperature : -50°C to +70°C
Opertaing Pressure : Atmospheric to 0.5 barg

ELECTRMAGNETIC COMPATIBILITY (EMC)

Immunity : Complies with EN 50082-1 : 1992
Emissions : Complies with EN 50081-1 : 1992

FLOW RATE TO SENSOR : 2 to 20 litres/minute

CONNECTIONS : *Swagelok* quick-connect stainless steel coupling and 0.25" overall diameter compression tube fitting.

WARRANTY : 2 years from date of delivery against faulty materials or workmanship.

Ordering Information

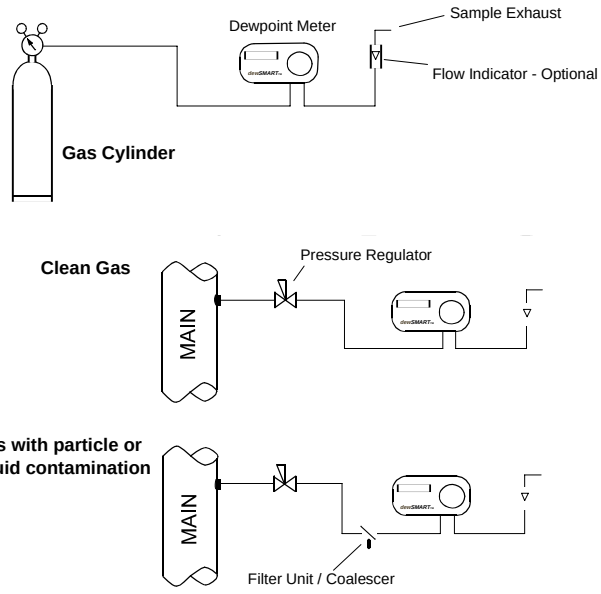
Model DSP-Rm- □□

CODE	RANGE
PL	-110 to +20°C
SR	-110 to -20°C
BL	-80 to +20°C
GY	-80 to 0°C
RD	-80 to -20°C
YW	-60 to 0°C
GD	-50 to +20°C
GN	-30 to +20°C

Example : If your requirement is for an Automatic Dewpoint Meter with a range of -80 to -20°C dewpoint then please order as : *Model DSP-Rm-RD*

TRANSIT CASE (OPTIONAL)
Order Code : Model DSP-TC

INSTALLATION SCHEMATICS

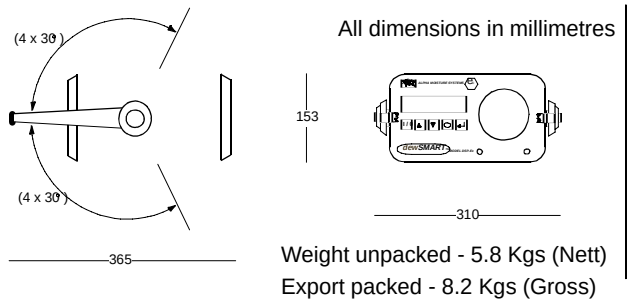


For details on additional accessories please see Accessories Specification sheet

Corrosive Gases

The Sensor should not be exposed to corrosive gases (or corrosive contaminants in the main gas sample) as they would chemically attack the sensor and render it useless. Examples of such gases are mercury (Hg), ammonia (NH_3), chlorine (Cl_2) and wet acid vapours i.e. acid vapours in gas with moisture content greater than 100ppm (v). Strong oxidising agents such as ozone (O_3) should also be prevented from coming into contact with the sensor.

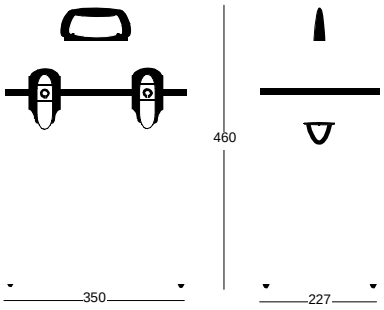
WEIGHTS & DIMENSIONS



All dimensions in millimetres

Weight unpacked - 5.8 Kgs (Nett)
Export packed - 8.2 Kgs (Gross)

TRANSIT CASE (Optional)



Weight unpacked
(Excluding analyser)
4.0 Kgs (Nett)

Export packed
(Excluding analyser)
6.0 Kgs (Gross)

Export packed
(Including analyser)
12.0 Kgs (Gross)

Product specification may be subject to change, without prior notice, as part of our ongoing product development programme.